Mathematical Foundations for Data Science & Computational Intelligence

All the Knobs at Once

Module 5 · Multivariable Calculus · Session 5.2

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Where We Left Off

Partial derivative

Nudge one knob, freeze the rest.

Tuesday's example

$$f(x, y) = x^2y + 3y$$
$$\partial f/\partial x = 2xy$$
$$\partial f/\partial y = x^2 + 3$$

Still open

~124 M

nudges per step for GPT-2 small. Today builds the language; the fix comes next Thursday.

Today: stop asking one knob at a time. Ask all of them at once, and get the answer as one object.

Today

1	Recap	5 min
2	The gradient, and which way is downhill	15 min
3	Directional derivative: why the gradient is steepest	15 min
4	The Jacobian: when the output is a vector	15 min
5	Chain rule: composition multiplies Jacobians	20 min
6	Where you'll meet this at work & wrap-up	10 min

By the end: you can differentiate a neural-network layer by hand.

Stack the Partials

$$\nabla f(x) = \left(\partial f/\partial x_1,\ \partial f/\partial x_2,\ \ldots,\ \partial f/\partial x_n \right)^T$$

A column vector with one entry per knob. GPT-2's gradient has 124 million entries.

Example

$f(x, y) = x^2y + 3y$

$\nabla f = (2xy,\ x^2 + 3)^T$

At (1, 1): $\nabla f = (2, 4)^T$

What the Gradient Is For

∇f

points in the direction of steepest ascent

$-\nabla f$

points steepest downhill

$\|\nabla f\|$

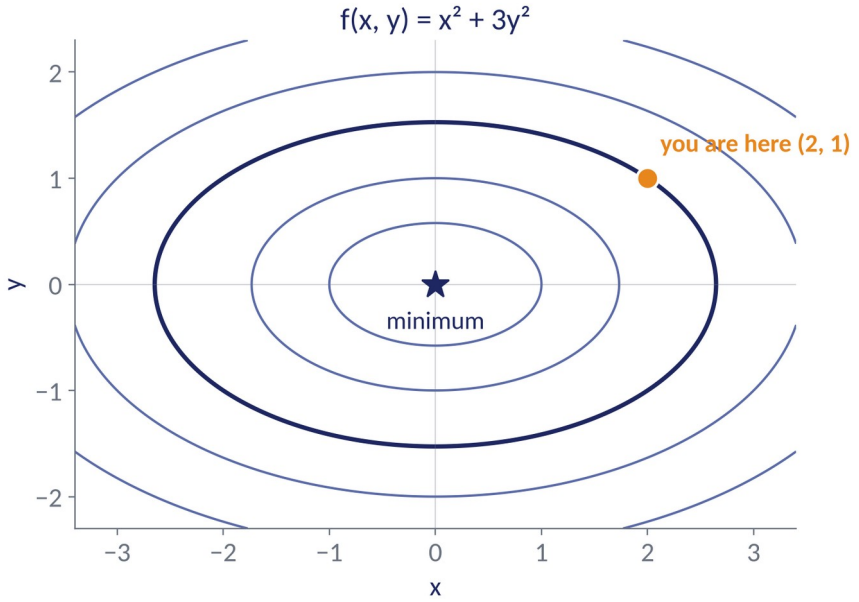
how steep it is

Training = take a small step in the direction $-\nabla L$. Repeat.

Today: the direction only. How far, and whether it converges: Module 8.

QUESTION FOR THE ROOM

I'm standing at the dot. Point in the direction of steepest descent.

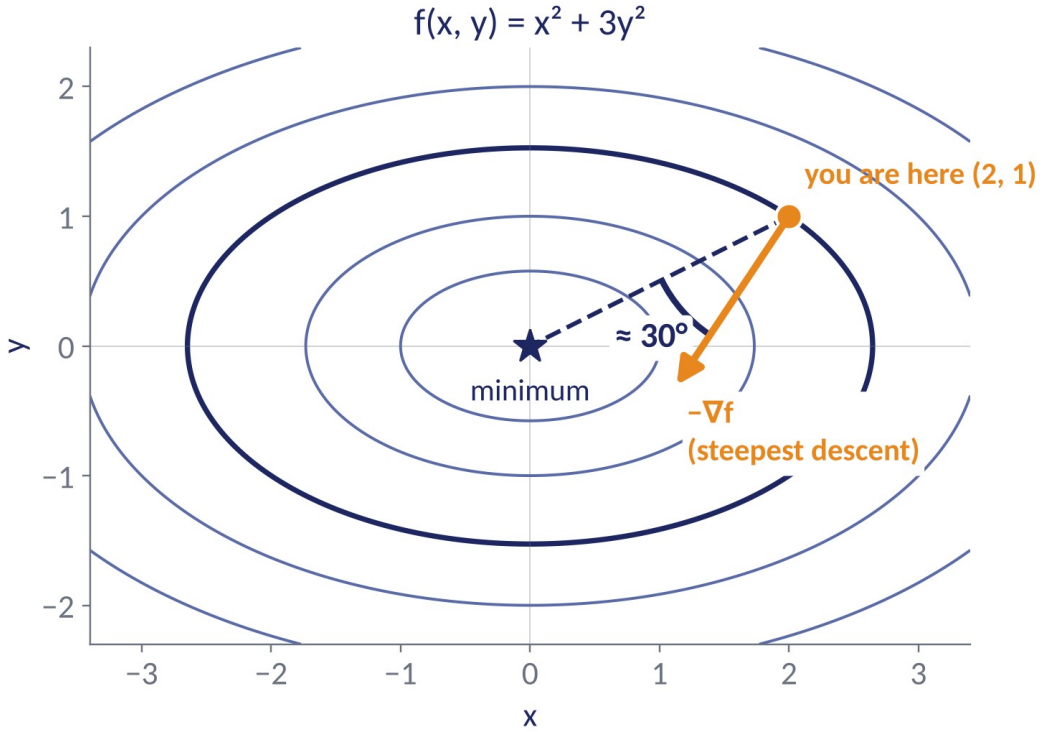


Compute It

1. **f(x, y) = x² + 3y²** $\nabla f = (2x,\ 6y)^T$
2. **At (2, 1)** $\nabla f = (4,\ 6)^T \quad \rightarrow \quad \text{downhill } -\nabla f = (-4,\ -6)^T$
3. **Straight to the minimum** $(-2,\ -1)^T$

Are (−4, −6) and (−2, −1) the same direction?

The Arrow Misses

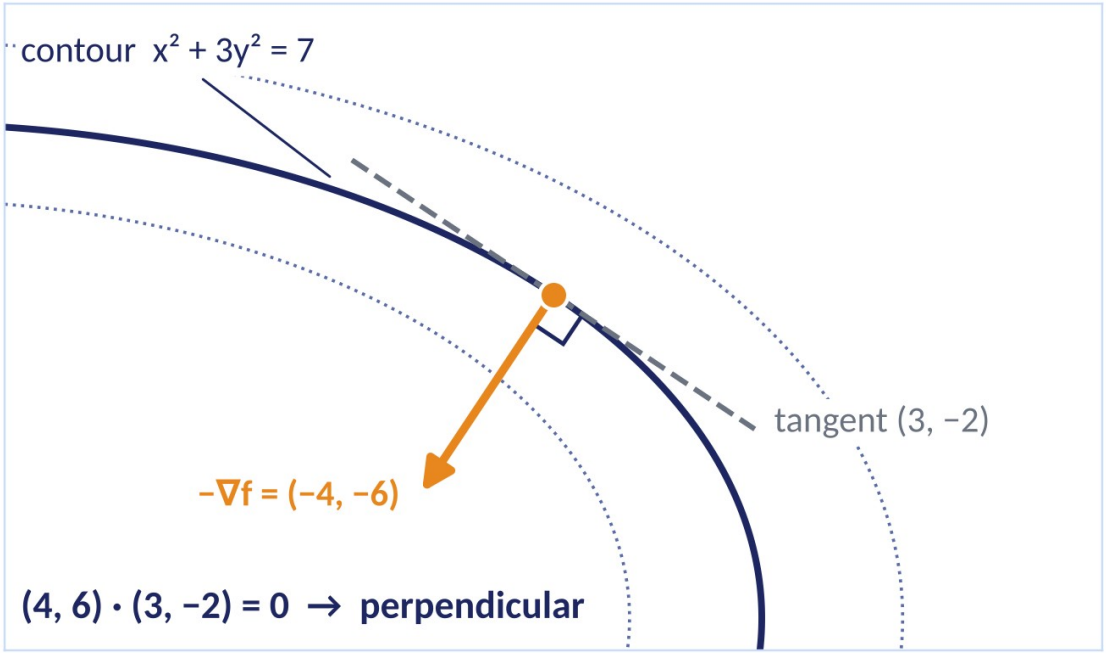


≈ 30°

cos θ = 14 / √260 ≈ 0.87
θ ≈ 29.7°

The steepest way down does not point at the bottom.

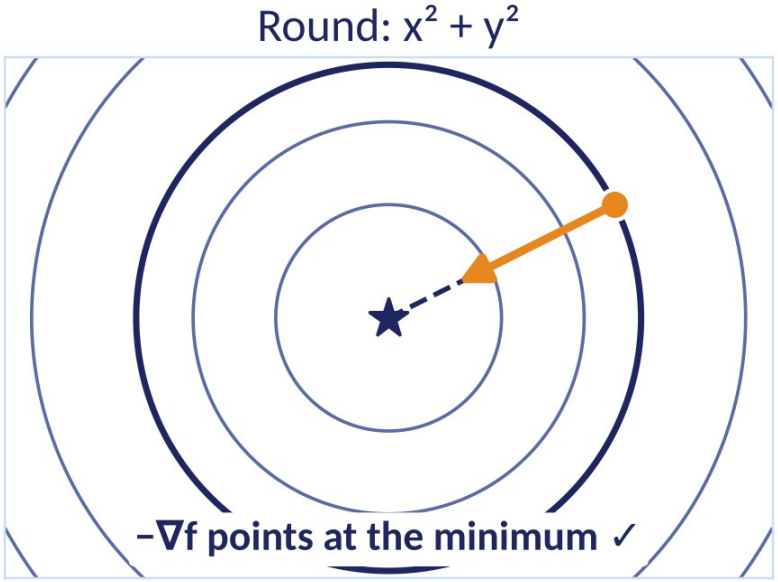
Perpendicular to the Contour

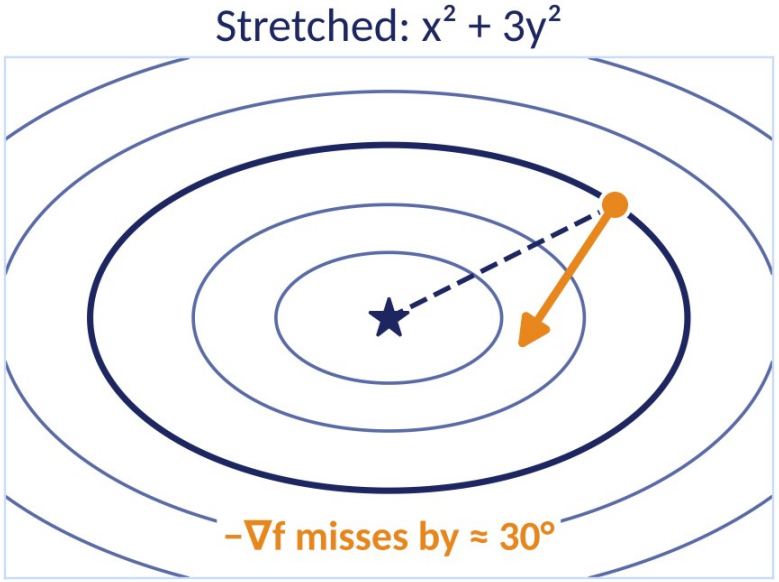


The gradient doesn't know where the minimum is.
It only knows the local slope, and the steepest local slope always runs straight across the contour lines.

Like water on a hillside: it doesn't aim for the lake.

Round Bowl vs. Stretched Bowl





What's different about the right bowl?

How Fast Does f Change in Direction u?

$$D_u f = \nabla f \cdot u \qquad (u \text{ a unit vector})$$

East

$u = (1, 0)$
$D_u f = \partial f / \partial x$

North

$u = (0, 1)$
$D_u f = \partial f / \partial y$

Any other direction

a mix of the partials

Partial derivatives are just the directional derivative along the axes.

Which Direction Is Steepest?

$f = x^2 + 3y^2$ at $(2, 1)$, $\nabla f = (4, 6)$

Direction	u	$D_u f$
East	(1, 0)	?
North	(0, 1)	?
Diagonal	(1, 1) / √2	?
Along ∇f	(4, 6) / √52	?
Along the contour	(3, −2) / √13	?

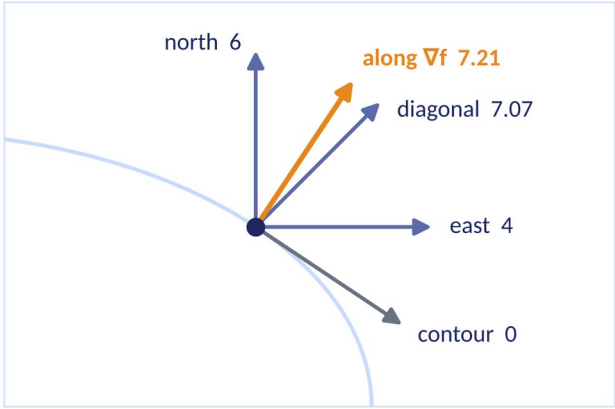
Rank these five before computing anything.

Then: can the diagonal beat 6?

Which Direction Is Steepest?

$f = x^2 + 3y^2$ at $(2, 1)$, $\nabla f = (4, 6)$

Direction	u	$D_u f$
East	(1, 0)	4
North	(0, 1)	6
Diagonal	(1, 1) / √2	≈ 7.07
Along ∇f	(4, 6) / √52	**≈ 7.21**
Along the contour	(3, −2) / √13	0



Nothing beats ‖∇f‖ = √52 ≈ 7.21.

The Proof Is a Dot Product

Rate of change in unit direction u: $D_u f = \nabla f \cdot u$

$$\nabla f \cdot u = \|\nabla f\| \, \|u\| \cos\theta = \|\nabla f\| \cos\theta$$

$\theta = 0°$

maximum, $\|\nabla f\|$
steepest ascent

$\theta = 90°$

zero
along the contour

$\theta = 180°$

minimum, $-\|\nabla f\|$
steepest descent

Quick Check · Pairs

$$f(x, y) = x^2y + 3y \quad \text{at } (1, 1): \quad \nabla f = (2, 4)$$

1. What's the rate of change heading east?
2. What's the maximum possible rate of change?
3. Name a direction where f doesn't change.

When the Output Is a Vector

$$f : \mathbb{R}^n \to \mathbb{R}^m$$

n inputs, m outputs

Jacobian J is **m × n**, with

$$J_{ij} = \partial f_i \,/\, \partial x_j$$

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} \begin{matrix} \leftarrow \text{output } 1 \\ \\ \leftarrow \text{output } m \end{matrix}$$

Rows are outputs. Columns are inputs.